

Evolution of Positionless Basketball and Associated Injury Risk in the National Basketball Association

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Abstract

The National Basketball Association (NBA) has increasingly shifted toward positionless basketball; however, there is no clear quantification of this shift. At the same time, there has been an increasing injury prevalence in the NBA, often attributed to workload, play style, and team-related factors. To evaluate when positionless basketball evolved in the NBA and examine potential associations with injury risk, we construct a positionless index using machine learning-based positional entropy from player-game data from 1982 to 2026. We then apply change-point detection and predictive modeling for injury risk within the following 14 days. We identify a mid-2010s structural increase in positionless basketball, with an 8.2% decrease in positional separability after this shift, as measured by mean pairwise centroid distance. A CatBoost classifier for injury prediction achieved an area under the curve (AUC) of 0.62 and a precision of 0.280, outperforming the baseline positive prevalence of 0.198. The classifier achieved a Brier calibration score of 0.158. Higher positionless index is associated with increased injury risk ($OR = 1.17$, 95% CI: 1.15-1.19) after adjusting for covariates. This relationship is nonlinear, with both positionally rigid and ambiguous players facing an increased injury risk. Additionally, this association is amplified by playmaking, perimeter defense, and spacing responsibilities. These findings support the use of positionless index as a quantifiable, continuous measure of role ambiguity in the NBA and suggest a need for increased monitoring of positional ambiguity for injury prevention strategies. Such risk mitigation strategies may involve managing positional ambiguity in player roles.

Keywords: Injury Risk | Basketball | Positional Versatility | Sports Medicine

1 Introduction

Teams in the National Basketball Association (NBA) have become increasingly positionless, where players play at various positions to suit distinct lineup styles. Prior research finds that role clusters, rather than player positions, may be better suited to describing players' styles and teams' compositions [1]. Further evidence for the positionless nature of the NBA comes from a machine learning (ML) model trained on player data from the 2021 season, which found a 71% accuracy in predicting players' positions [2]. However, few studies have quantified the degree of positionless play per player. One such study created a binary indicator of positionlessness based on ML models' predicted position probabilities [3]. Other recent work has quantified variability in play style, showing that versatility can be demonstrated beyond the traditional five position labels [4]. Still, positionlessness, rather than lineup-dependent role adaptation, has not been defined as a continuous metric. Additionally, the timing and magnitude of the NBA's trend toward positionless basketball is unresolved.

Prior work has found that positionless players, or those with more positional ambiguity, seem to have more role conflict and associated mental stress and cognitive overload [5], which could lead to injury susceptibility. Notably, the NBA has had an increasing prevalence of injuries in recent years [6].

This increased injury rate has, in part, led to an open debate surrounding load management practices, where teams rest players to prevent injuries. However, a ten-year, NBA-commissioned epidemiological study found no link between load management and reduced injuries [7]. On the other hand, high playing time, defined as more than 100

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minutes over a 5-game stretch, seems to correlate with increased injuries, especially muscle strains and contusions, in a deep learning model [8]. Separate studies found workload, measured both by minutes and usage rate, to correlate with injury risk [9, 10]. Other factors predictive of injury risk have included physical characteristics, such as height and weight, [11] and play style, using free throw and 3-point attempt rates as proxies [12]. However, the relationship between positionless basketball and injury susceptibility has not been well-characterized.

Therefore, we examine when the NBA became more positionless and create a quantitative positionless index for NBA players. We then determine if this positionless index correlates to increased injury risk while adjusting for other factors, including workload, play style, and team context.

Data

Data Sources

This study combines several datasets. For player and team statistics, box scores from NBA regular season, playoff, and play-in games were analyzed from the 1982-83 season through the 2025-26 season [13]. Individual player games were analyzed for all players logged at least 10 minutes and one field goal attempt, and each player-game was merged with team-game statistics for that player’s team. To find player positions, Basketball Reference season-level tables were scraped [14]. The NBA Stats API was used for players’ biographical information and individual game logs, which were used to generate travel data [15].

Finally, a list of NBA injuries from 2011 onwards was scraped from Hashtag Basketball’s NBA Injury Database [16]. Since this database does not provide a full list of injuries, a seed-based scraping algorithm was implemented. This algorithm begins with keywords from the database’s ten most common injuries; after searching these terms, the results are split into tokens, which are again searched. This iterative process yielded 727 distinct injury types and 8,425 player injuries.

Data Preprocessing

Players with a missing amount of minutes or fewer than 10 minutes played were dropped from the analysis, excluding 521,838 player-game observations (31.3% of the full dataset). An additional 8,402 (0.7% of the remaining data) player-game observations did not contain an attempted field goal and were excluded due to the calculation of field goal percentage. 230,534 (20.3%) player-games occurred before the 1982-83 season and were dropped due to the absence of team statistics, such as total field goals attempted. 33,351 player-game observations (3.7%) were dropped due to missing team statistics after the beginning of the 1983-84 season. 88 player-game observations (0.0001%) were dropped from the analysis due to a position (e.g., F) outside of the conventional five positions. The final dataset was approximately evenly balanced between positions, with most player-game observations representing shooting guards (21.3%) and the fewest representing centers (19.0%). Point guards, small forwards, and power forwards comprised 20.0%, 19.3%, and 20.4% of the dataset, respectively. Median imputation was used for variables with minimal missingness not at random.

Feature Engineering

Per-36 statistics and percentages relative to the total team statistic (e.g., percent of team’s points scored by that player) were calculated. Relative shooting percentages, defined as the team’s shooting percentage subtracted from that for the player, were calculated for field goals, three-point attempts, and free throws. Free throw and three-point attempt rates, alongside true shooting, effective field goal, and usage rates, were calculated.

From travel data, games played in the last 14 days was calculated. For injury prediction, 3- and 7-game rolling statistics were generated to avoid data leakage from a game during which the player was injured.

Methods

Position Prediction

To evaluate the rise of positionless basketball in the NBA, season-level models were constructed to predict players’ positions. Positionlessness was defined as the inability of these models to accurately predict player positions.

To reduce multicollinearity, highly correlated features were dropped using a networks-based feature reduction algorithm. All features with a Pearson correlation greater than or equal to 0.8 were used to construct a correlation

network. Iteratively, the most representative features, defined as the node with the most uncovered neighbors, was selected, with ties broken using degree centrality. This process continued until all features in the network were either selected or correlated with a selected feature. Boruta selection was then run on the remaining features using data from all seasons, selecting 21 for season-level player position prediction.

Various models were compared without hyperparameter tuning, of which LightGBM Classifiers performed the best. Five seasons, evenly spaced at ten-year intervals, were subsequently used for hyperparameter tuning. 80% of the data was used for training, and the split between training and testing was stratified by position. A randomized search of optimal hyperparameter configurations was conducted, and these hyperparameters were utilized for each season. Area under the curve (AUC) for player position prediction was calculated for each season, and change-point detection using Pruned Exact Linear Time (PELT) was performed [17]. The penalty β was modulated in sensitivity analyses. For additional sensitivity analyses, PELT was applied to the season-level time series of mean pairwise Euclidean distance between position centroids. Position centroids were computed as the mean of globally standardized feature vectors within each position and season. The first two seasons were excluded due to noise caused by low sample size.

Positionless Index

The positionless index for each player was calculated as the entropy of out-of-fold position probabilities for each player-game observation for a particular season using 5-fold stratified cross-validation. Out-of-fold probabilities were used to avoid optimistic bias from evaluating positional uncertainty on data used for training. Entropy ranged from 0 to 2.32, with increasing entropy denoting increasing positional ambiguity. The final index for each player-season was defined as the average entropy across folds. Since models were trained at a season-level and entropy was calculated at this level, increased entropy represents positional ambiguity rather than predictive uncertainty inherent to the model.

Equations 1 and 2 define this calculation, where i indexes player-game observations, p represents players, s represents seasons, and k represents the five position classes. p_{ik}^{OOF} denotes the out-of-fold probability that player-game i plays position k , $G_{p,s}$ denotes all player-game observations for player p in season s , and $N_{p,s}$ is the total number of player-game observations in $G_{p,s}$.

$$H_i = - \sum_{k=1}^5 p_{ik}^{\text{OOF}} \log_2 p_{ik}^{\text{OOF}}$$

$$PI_{p,s} = \frac{1}{N_{p,s}} \sum_{i \in G_{p,s}} H_i$$

Injury Risk Prediction

For injury risk prediction, data was limited to injuries occurring between January 4, 2011 and May 13, 2025 due to database limitations. A binary indicator of injury occurrence within 14 days of the player-game observation was constructed and used as the target variable for prediction. Since the target variable is binary, multiple injuries in the window counted once. Sensitivity analyses were conducted using binary indicators within 7 and 30 days of player-game observations.

Injury risk prediction was adjusted using team context, rolling game-level statistics, travel information, and biographical player information. Team context included if the game was a win or at home as well as the team’s overall win percentage that season, and travel information included flight distance from the last day, timezone shift, and rest days. Height, weight, body mass index (BMI), age, and years of experience were used as biographical player information.

To reduce the number of features, features that did not achieve a univariate logistic regression AUC of 0.52 were excluded from the analysis, removing variables exhibiting negligible discriminatory ability. Highly correlated features were pruned using the previously described networks-based feature reduction algorithm. Overall, 34 were kept.

Logistic regression, random forest, XGBoost, LightGBM, CatBoost, and multi-layer perceptrons were compared without hyperparameter tuning. Class imbalance was addressed using inverse-frequency class weighting, but this adjustment did not materially change AUC across models. Therefore, unweighted models were constructed for these analyses. Each model was trained on data until 2022-23, with 2023-24 data used for validation. The training dataset

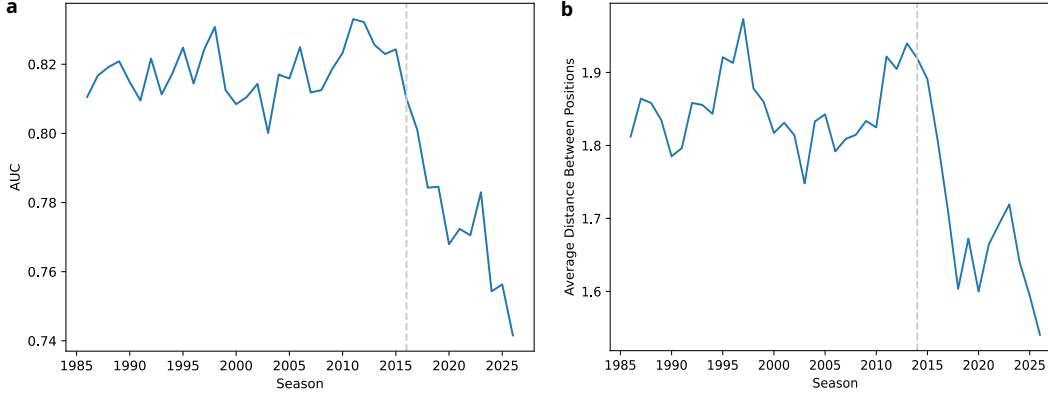


Figure 1: **Positionless basketball changed around 2014-2016.** (a) Decrease in area under the curve (AUC) in player position prediction over time. (b) Decrease in position heterogeneity, measured by average pairwise Euclidean distance between position centroids, over time.

included 286,926 player-game observations with a 9.3% injury incidence rate, while the validation dataset included 23,295 observations with an 18.8% incidence rate. A CatBoost classifier performed the best and was trained on the combined training and validation datasets. AUC, average precision, and precision at top-k percentiles were used to evaluate model performance on 2024-25 season data. This test data included 22,937 player-game observations with a 19.8% injury incidence rate. The Brier calibration score was also calculated [18]. SHAP values were used to evaluate feature importance for the CatBoost classifier [19]. A logistic regression was also run as a sensitivity analysis.

To understand covariates' impacts on the association of positionless index and injury risk, a logistic regression consisting of interaction terms was constructed. Odds ratios are reported with 95% confidence intervals. p-values were adjusted for multiple comparisons using the Benjamini-Hochberg procedure. Statistical significance was defined as an adjusted p-value less than 0.05.

Results

To examine the onset of positionless basketball, change-point detection was applied to season-level model performance and pairwise Euclidean distance between position centroids. Season-level model performance in player position prediction diminished in 2016 (Figure 1a). Consistent with 2016 as the change point, mean position-prediction performance declined from an average AUC of 0.8181 in 1986-2015 to 0.7751 in 2016-2026, representing a relative decline of 5.3%. Similarly, the pairwise Euclidean distance between position centroids decreased in 2014 (Figure 1b) according to change-point detection, indicating that positions became more similar. The mean pairwise Euclidean distance between position centroids across 1986-2013 was 1.849, while this decreased by 8.2% to 1.697 across 2014-2026.

Having demonstrated validity of the positionless quantification construct compared to pairwise position distance, each player was assigned a positionless index. A CatBoost classifier predicting injury in the 14 days following a player-game observation achieved an AUC of 0.620. The average precision score of 0.280 exceeded the baseline positive class prevalence in the test set of 0.198. Injury risk increased with predicted risk percentile. Precision at the top 1%, 5%, 10%, and 20% of samples by predicted probability was 0.410, 0.372, 0.338, and 0.300, respectively. The observed injury rate of 0.410 in the top 1% percentile of predicted injury risk probability corresponded to a 2.07-fold enrichment relative to the baseline prevalence of 0.198.

In the CatBoost classifier, positionless index achieved the highest feature importance among features related to workload, play style, and team context such as travel. Figure 2a demonstrates that players with a higher positionless index had a higher chance of being injured within the next 14 days. Other important features included teams' season win percentage and players' play styles, including three-point attempts, blocks, assists, and rebounds. Seven-game rolling minutes and games played over the prior 14 days also contributed and represented player workload. Sensitivity analyses using 7-day and 30-day time horizons achieved similar results, finding positionless index to have the highest SHAP feature importance among the same feature set within the CatBoost classifier.

Figure 2b adjusts for similar factors and finds the highest odds ratios for seven-game rolling statistics involving points per 36 minutes and three-point attempts. Still, higher values of positionless index were associated with

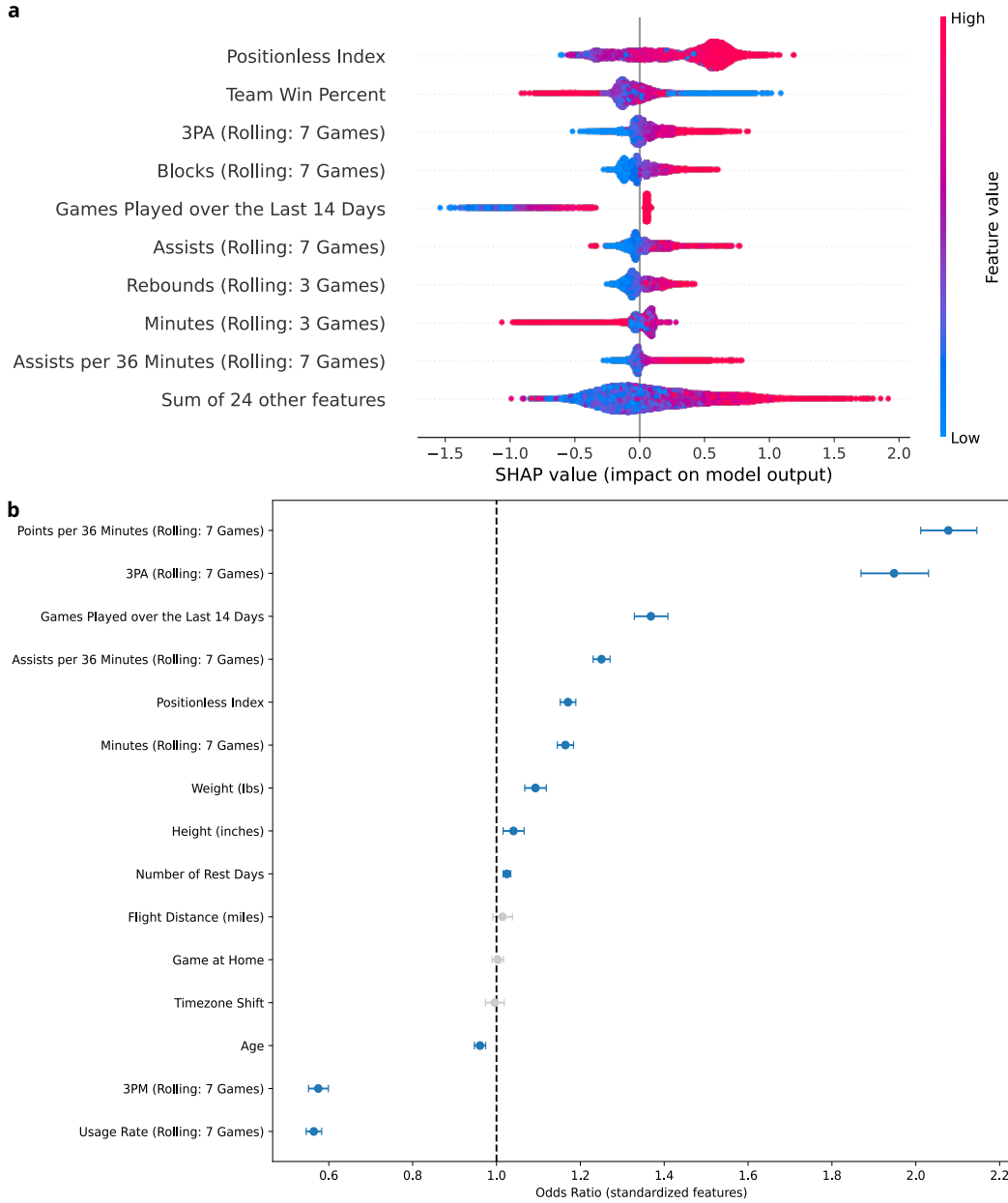


Figure 2: **Positionless index significantly predicts injury risk after adjusting for workload, play style, and team context.** (a) SHAP values for a CatBoost Classifier. (b) Forest plot for a logistic regression.

increased odds of injury within 14 days (OR = 1.17, 95% CI: 1.15-1.19, Adjusted p-value < 0.0001).

A SHAP dependence plot of positionless index in predicting injury risk (Figure 3a) calculated on the 2024-25 season approximates a U-shaped curve, supporting a nonlinear relationship between these variables. Low and high values of positionless index were associated with increased injury risk, while intermediate positionless play styles were associated with a reduced risk. This pattern attenuated at the extreme lower end of the positionless index distribution although this deviation may be driven by sparse observations. The average injury rate within 14 days of the player-game observation by positionless index decile in the whole dataset (Figure 3b) supports this observation. Players in the lowest decile of positionless index experienced a greater injury rate (11.83%) than players in the next six deciles (mean injury rate of 9.18%). This effect remained in a sensitivity analysis adjusting for workload (average minutes played in the prior seven games). When adjusting for workload, the lowest decile of positionless index was associated with an injury rate of 16.48% per 36 minutes, while the next six deciles were associated with an injury rate of 11.87%. The final three deciles were associated with an average injury rate of 20.23% per 36 minutes played on average across the previous seven games.

To examine what variables influence the association between positionless index and injury risk, interaction effects of covariates related to play style, workload, and team context were evaluated (Figure 4). Play style-related

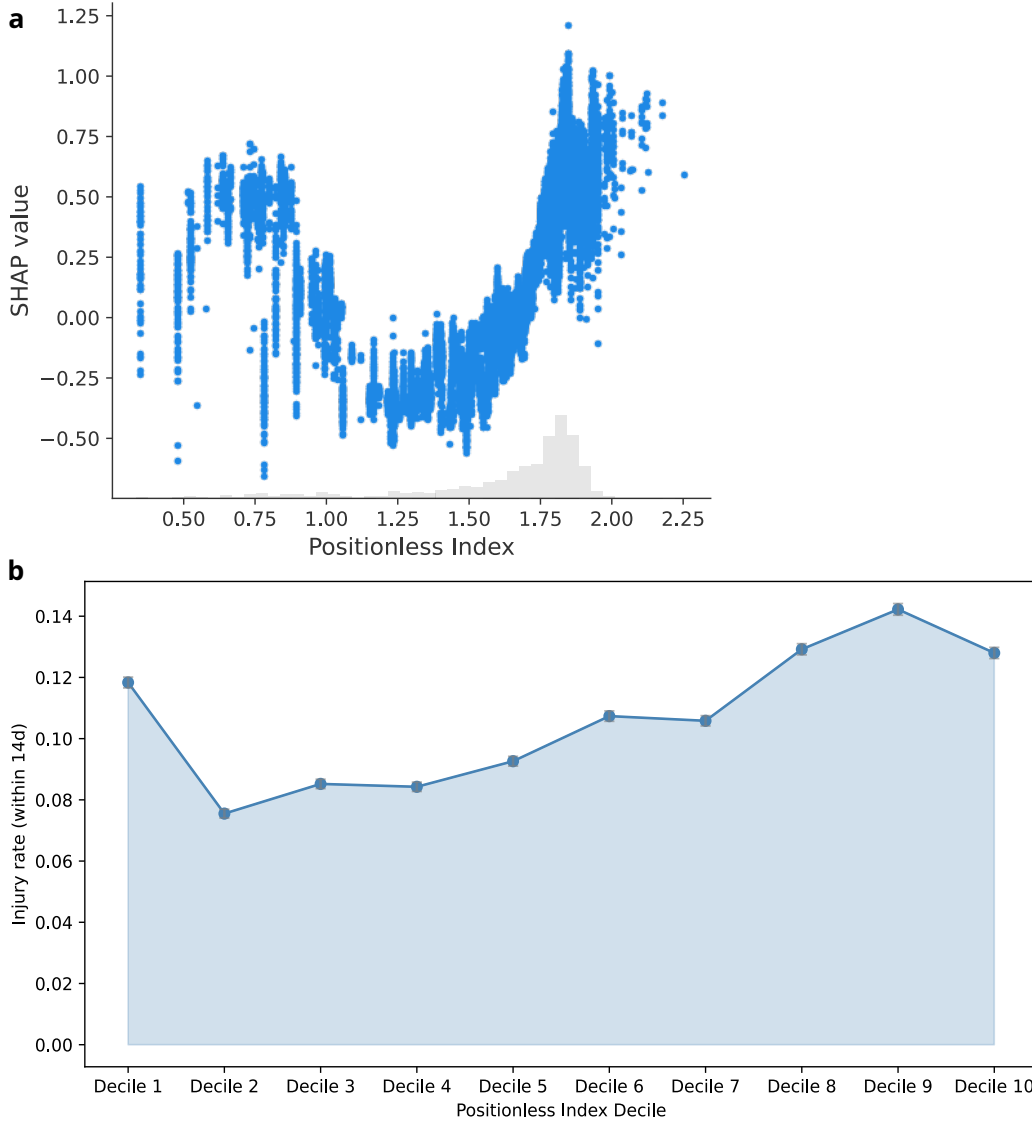


Figure 3: Positionless index and injury risk exhibit a nonlinear relationship. (a) SHAP dependence scatter plot of positionless index and SHAP values in injury risk prediction. (b) Average injury risk within the next 14 days based on positionless index decile.

variables, such as spacing responsibilities, playmaking, and perimeter defense, amplified the effect of positionless index on injury risk. Specifically, the average number of three pointers attempted over the prior seven games had the strongest interaction effect on the measured association (OR = 1.18, 95% CI: 1.10-1.27, Adjusted p-value < 0.0001). Playmaking, measured by the average number of assists over the prior seven games, had the second strongest interaction effect (OR = 1.17, 95% CI: 1.09-1.25, Adjusted p-value < 0.0001). Steals, representative of perimeter defense, also amplified the relationship between positionless index and injury risk (OR = 1.06, 95% CI: 1.01-1.10, Adjusted p-value = 0.03).

Discussion

Our findings demonstrate that basketball in the NBA is increasingly positionless. The shift toward positionless basketball is quantifiable and began around 2014-2016. Players with positionless play styles tend to have increased injury risk, and this association is amplified by playmaking, perimeter defense, and spacing responsibilities.

This study creates a novel and continuous variable quantifying positionless basketball in the NBA. Shawn Marion, a four-time NBA All-Star, placed positionless basketball's start in the early 2010s [20], and our results find that overall league-wide adoption came in the middle of the decade. Additionally, a change-point detection analysis without positionless index found a similar shift (Figure 1b). Future research may use positionless index as a novel method to quantify this trend. Additionally, more datasets, including those covering distance covered

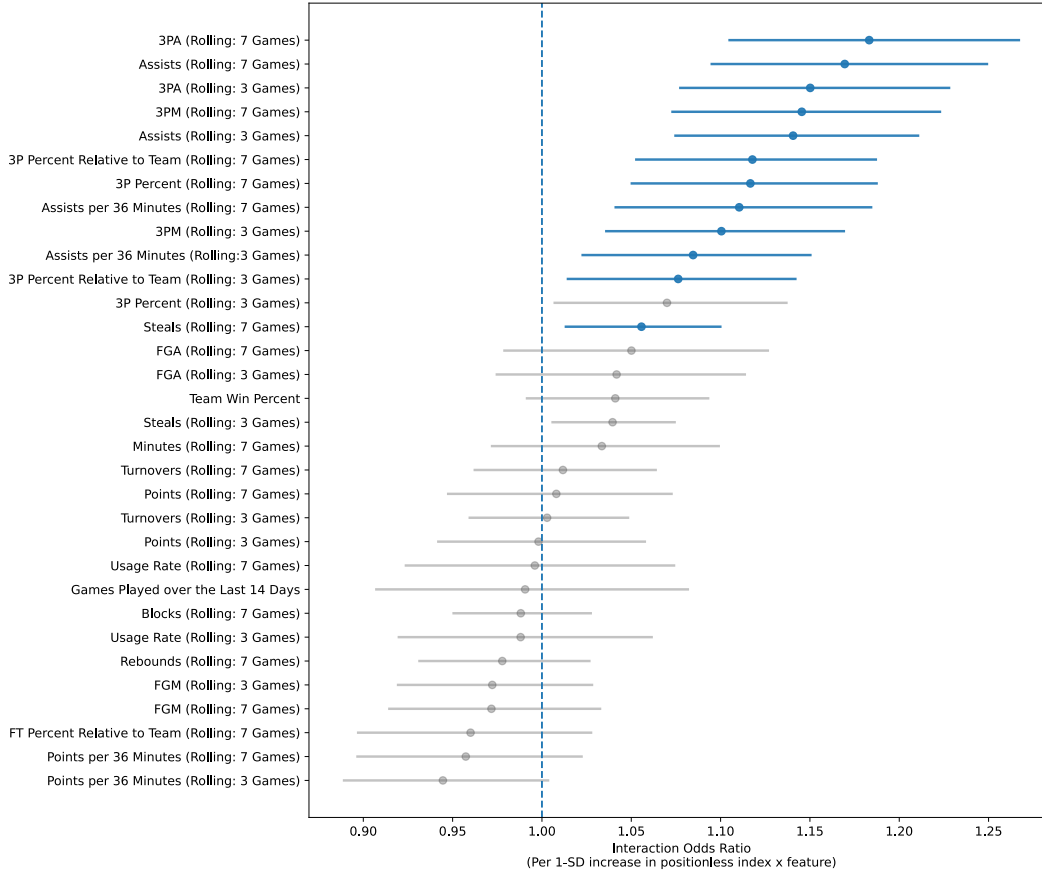


Figure 4: **Play style-related variables amplify the positionless index’s effect on injury risk.** Increased 3-point attempts, makes, and percentages; assists; and steals amplify this interaction.

and defensive assignments, can be leveraged to improve position prediction. Adding more features can both further validate positionless index as an analytical metric and lead to more precise entropy calculations.

Increased positionless play is associated with greater injury risk. Previous studies predicting injuries in the NBA have analyzed workload, physical characteristics such as height and weight, and play styles [8, 11, 12]. Our results analyzing these variables agree with prior studies, finding increased injury risk associated with increased height and weight and more minutes played over the last seven games. However, positionless basketball has not been previously quantified or analyzed as a predictor of injury risk. Our results find positionless index to be a significant covariate in injury prediction across multiple methods and time horizons.

The association between positionless index and injury risk is nonlinear. Moderate role specialization seems the least injury-prone, while highly rigid and highly ambiguous roles are both associated with increased injury risk. Highly rigid roles, such as the traditional center protecting and attacking the rim, may lead to increased wear and tear. However, players with ambiguous positions may face increased role strain and the resulting cognitive and physical demands [5]. Still, a positionless player’s skillset matters: high-intensity offensive or defensive responsibilities such as high shooting volume, playmaking responsibilities, or steals amplify the injury risk due to positionless basketball.

These findings have several implications for NBA teams and players. First, we recommend monitoring positional entropy as a screening metric for future injury risk. Next, we recommend that NBA teams manage positionless players’ skillsets to avoid concentrating perimeter defense, play creation, and spacing responsibilities. These variables each amplify the increased injury risk due to positionless play styles. Together, these responsibilities create a high-risk environment for players. Additionally, teams should monitor or constrain positionless players’ responsibilities to those of fewer positions. Such constraints may be especially important in the regular season, where teams increasingly try to manage and prevent injuries. While this paper cannot causally compare these recommendations against the status quo of load management, this paper’s findings suggest that positionless players’ responsibilities affect injury risk at a similar rate to minutes played (Figure 2b). Therefore, following these recommendations while reducing load management practices could improve fan satisfaction with their team and align with NBA goals of decreasing this practice [21] while ameliorating injury rates. Still, we do not recommend an end to load management practices due to our results’ agreement with those of other studies finding that increased minutes correlate with

increased injury risk [8, 9].

Beyond injury prediction and management, positionless index may provide a new framework to study roster construction and player development. This index may offer a complementary perspective to the traditional five position labels, adding new information to the breadth of particular players’ on-the-court responsibilities.

Limitations

Since the injury database utilized in this study does not provide a full list of injuries, the list used is incomplete but rather contains many of the most common NBA injuries. Moreover, modern injury reporting may be more granular, and different teams make distinct decisions on whether to play players through injury. The list of injuries used does not include data from before 2011. Since this is an observational study, we cannot establish causality from our findings. Our calculation of positionless index may be sensitive to feature selection and the models used. This calculation was dependent on Basketball Reference, which assigned a single position label per season. However, some players play different positions within the season. The injury risk predictive model achieved an AUC of 0.62, which outperformed baseline injury prevalence but should be considered limited predictive accuracy. Finally, this study utilized NBA data and may not be generalizable to the WNBA or other basketball leagues. Still, this is the first article to quantify positionless basketball in the NBA and demonstrate its association with injury risk, and we use multiple performance metrics and sensitivity analyses while avoiding data leakage using a temporal split.

Code and Data Availability

The code used for this paper’s analysis can be found at <https://github.com/vedanttapiavala/positionless-NBA-analytics>. A website including player-season’s positionless indices and players’ positionless index trajectories by season is available at <https://positionless-nba-analytics.vercel.app/>.

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